

Two Eight-Element Distributive Lattices: A Disproof of Conjecture 00000002604

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Abstract

The source asserts that finite distributive lattices with the same layer-count vector are isomorphic. The lattices $C_4 \times C_2$ and $(C_3 \times C_3) \setminus \{(2, 0)\}$ are distributive, have eight elements each, and have layer-count vector $(1, 2, 2, 2, 1)$ whether layers are counted from the bottom or from the top. They are not isomorphic. All of this, and the negation of the universal claim, is verified in Lean. An earlier submission used an equivalent example but did not establish non-isomorphism; its error is described below.

Reading of the source. A finite distributive lattice is graded: it has a unique rank function ρ with $\rho(\perp) = 0$ and $\rho(b) = \rho(a) + 1$ whenever b covers a . Its k -th layer is $\{x : \rho(x) = k\}$. The English text speaks of the layer-count vector; the Chinese text counts the layers from the coatoms downwards, that is by the corank $\rho(\top) - \rho(x)$. We treat both. Layer 1 of the corank is the set of coatoms.

Theorem 1. *Let C_n be the chain $0 < 1 < \dots < n - 1$, and let*

$$L_1 = C_4 \times C_2, \quad L_2 = \{(i, j) \in C_3 \times C_3 : (i, j) \neq (2, 0)\},$$

both with the componentwise order. Then L_1 and L_2 are distributive lattices with eight elements. Their layer-count vectors, by rank and by corank, are all equal to $(1, 2, 2, 2, 1)$. The lattices L_1 and L_2 are not isomorphic.

Proof. Lattices. A product of chains is a distributive lattice under componentwise maximum and minimum. Let $p, q \in L_2$. If $p \vee q$ has first coordinate 2, then p or q has first coordinate 2 and so a nonzero second coordinate; hence $p \vee q$ has a nonzero second coordinate. If $p \wedge q$ has first coordinate 2, then both p and q do, both second coordinates are nonzero, and so is their minimum. So L_2 is a sublattice of $C_3 \times C_3$ and is distributive.

Layers. In both lattices $\rho(i, j) = i + j$. This is clear for L_1 . For L_2 , every pair $a < b$ of L_2 that has $(2, 0)$ strictly between also has $(1, 0)$ or $(1, 1)$ strictly between. Removing $(2, 0)$ therefore creates no new covering relation, and every covering relation of L_2 raises $i + j$ by one. The layers are

$i + j$	0	1	2	3	4
L_1	(0, 0)	(1, 0), (0, 1)	(2, 0), (1, 1)	(3, 0), (2, 1)	(3, 1)
L_2	(0, 0)	(1, 0), (0, 1)	(1, 1), (0, 2)	(2, 1), (1, 2)	(2, 2)

so both vectors are $(1, 2, 2, 2, 1)$. This vector is a palindrome and the top element has rank 4 in both lattices, so the corank layers give the same vector. In particular both lattices have two coatoms.

Non-isomorphism. Call an element a *branch point* if two incomparable elements lie below it and two incomparable elements lie above it. This property is preserved by order isomorphisms. In L_2 the element $(1, 1)$ is a branch point: $(1, 0)$ and $(0, 1)$ lie below it, and $(2, 1)$ and $(1, 2)$ lie above it. In L_1 the elements below $(i, 0)$ form a chain, and the elements above $(i, 1)$ form a chain. So L_1 has no branch point, and no isomorphism $L_1 \rightarrow L_2$ exists. $\square$

Remark. By Birkhoff’s theorem, L_1 is the lattice of order ideals of the poset P formed by a three-element chain and an isolated point, and L_2 is that of the poset Q on $\{a, b, c, d\}$ with $a < c$, $b < c$, $b < d$. For Q the isomorphism sends an ideal I to $(|I \cap \{a, c\}|, |I \cap \{b, d\}|)$; the excluded pair $(2, 0)$ would be an ideal containing c but not b . This remark is not formalised and is not needed.

The earlier submission and its error

Pull request 48 of the competition repository (by `earthking11`, merged on 1 October 2026) treated this conjecture with the posets P and Q of the remark. The organisers removed it in the re-audit commit `541cf4f` of 3 October 2026, with the note that non-isomorphism was never established.

Its mathematics is sound: the two ideal lattices do have equal vectors and are not isomorphic. The error is in the Lean file. Its final theorem asserts

```
perms.all (fun p => IsIsoPerm leP leQ p) = false.
```

This says that it is false that all 24 bijections are isomorphisms, in other words that at least one bijection is not an isomorphism. It does not say that no bijection is an isomorphism, which is what non-isomorphism means. In addition, the file works with Boolean relations on four points and with bitmask counts of ideals. It never forms the two lattices, their order, or a statement about lattice isomorphism.

The present submission forms both lattices as Mathlib distributive lattices and proves that the type of order isomorphisms between them is empty.

Formal verification and scope

`L1` is the product type of `Fin 4` and `Fin 2` with Mathlib’s product order, which carries Mathlib’s `DistribLattice` instance. `L2` is the subtype of the product of `Fin 3` with itself defined by the condition above. Closure under supremum and infimum is proved, and the `DistribLattice` instance of the subtype is derived from it, so its operations are those of the ambient lattice. Both carry bounded orders.

`IsRankFunction` and `IsCorankFunction` are stated with Mathlib’s covering relation. The functions $i + j$ and $4 - (i + j)$ are proved to satisfy them on both lattices. The eight equalities of layer counts are kernel-checked, and `layer_rank_eq` and `layer_corank_eq` give equality of the layer counts for every k . `not_isomorphic` proves that the type of order isomorphisms from `L1` to `L2` is empty, by the branch-point argument.

The proposition

```
ClaimedCoatomLayerRigidity
```

quantifies over all pairs of finite bounded distributive lattices with corank functions and equal layer counts and asserts an order isomorphism. Its negation is `conjecture_false`. The same statement for rank functions is refuted in `rank_rigidity_false`. No `sorry`, `native_decide` or custom axiom occurs.

The disproof concerns the first clause of the source. Since two non-isomorphic lattices share a vector, no characterisation of the realisable vectors can repair that clause, and no claim is made about the Kruskal–Katona clause.

Source and reproduction

The exact bilingual source is included byte-for-byte as `SOURCE.md`. Its repository path is

`conjectures/00000002604.md`.

The project pins Lean 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

From `lean/`, run `lake build`, then

`lake env lean -DwarningAsError=true Main.lean`.

The included public-Git manifest locks all dependency revisions. The `verification/` directory records the fresh-build logs, theorem-axiom reports, artifact hashes, review and final PDF inspection. No supplementary numerical computation is required.